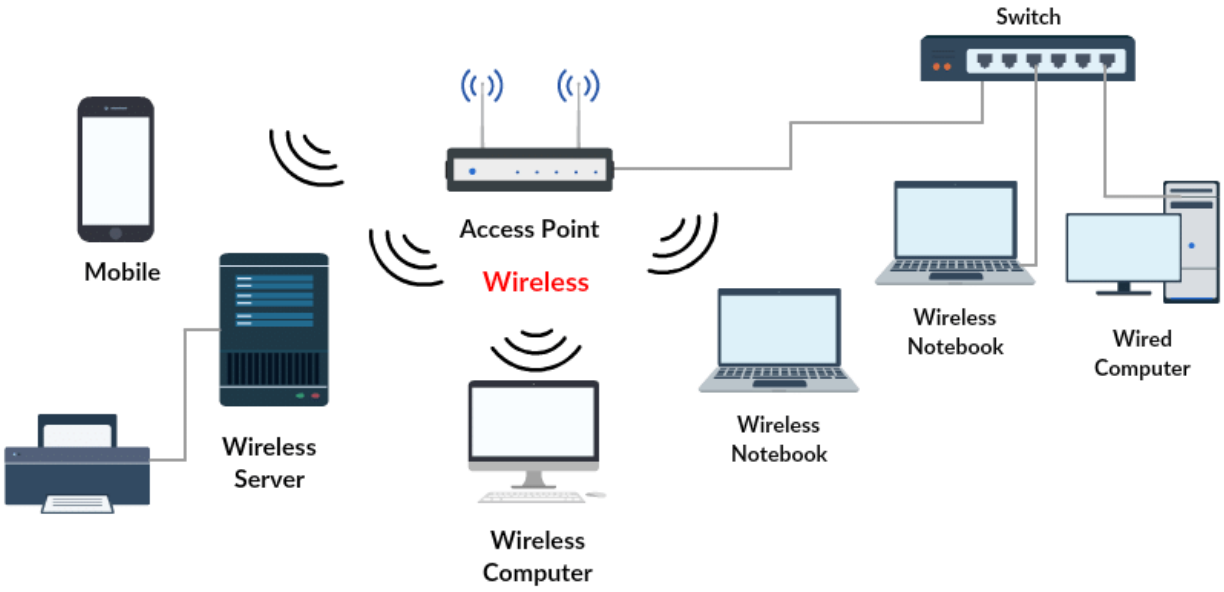
MOBILE COMMUNICATION ROUTING

Mobile computing is one of the most important technologies supporting pervasive computing. Advances in both hardware and software techniques have enabled the spread of mobile hosts and wireless networking to masses. Generally, there are two modes in which wireless mobile nodes can communicate:

1. **Infrastructured**: In this mode, all the communication among the mobile nodes goes through a base station. A Base station is also known as an access point. These base stations are connected to the fixed infrastructure or wired networks. This structure provides better reliability, security, and scalability compared to ad-hoc networks, enabling access to wired networks.



Infrastructured wireless networks

Key Components & Concepts

- **Access Point (AP):** Functions as a central bridge, connecting wireless clients to a wired network.
- **Wireless Router:** Integrates router and AP functions, common in home and office environments.
- **Hardware Components:** Include antennas, switches, and controllers for managing connectivity.
- **Network Modes:** In this mode, all traffic flows through the AP to manage traffic efficiently.

Advantages of Infrastructure Wireless

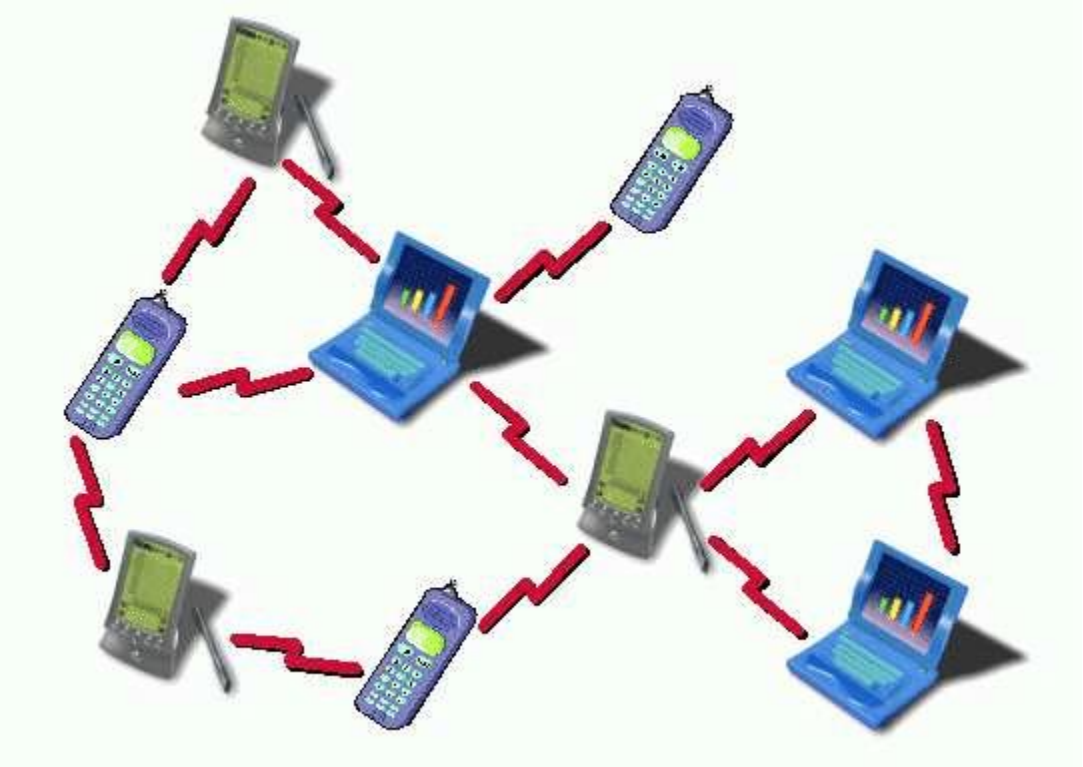
- **Scalability:** Allows adding more APs for increased coverage.
- **Security:** Supports robust security protocols like WPA3, often managed through centralized controllers.
- **Performance:** Generally provides faster, more reliable connections than direct device-to-device communication.

Applications

- **Corporate WLAN:** Large-scale deployment for business internet access.
- **Cellular Networks:** Base stations serving mobile users.
- **IoT Infrastructure:** Connecting sensors and devices within a building

2. **Infrastructureless**: This mode of communication is known as a mobile ad hoc network (MANET). A MANET is a collection of wireless nodes that can dynamically form a network to exchange information without using any pre-existing fixed network infrastructure.

   A typical example of this mode of communication is people sitting in the conference room and exchanging data among them without any fixed infrastructure.



Infrastructure less wireless networks/ Mobile ad hoc network

Mobile ad hoc network (MANET) is an autonomous system of mobile nodes connected by wireless links.

MANET is a collection of wireless nodes that can dynamically form a network to exchange information without using any pre-existing fixed network infrastructure.

In a MANET, each node or device is expected to serve as a router, and each router is indistinguishable from another in the sense that all routers execute the same routing algorithm to compute paths through the entire network.

Each node operates not only as an end system, but also as a router to forward packets. The nodes are free to move about and organize themselves into a network. These nodes change position frequently. To accommodate the changing topology special routing algorithms are needed.

There is no single protocol that fits all networks perfectly. The protocols have to be chosen according to network characteristics, such as density, size and the mobility of the nodes.

MANET does not require any fixed infrastructure, such as a base station, therefore, it is an attractive option for connecting devices quickly and spontaneously.

Mobile ad hoc networks can be used alone (for example in the military) or as a hybrid together with the Internet or other networks. Different MANET applications have different needs, and hence the various Mobile Ad Hoc Network routing protocols may be suitable in different areas. The size of the network and the frequency of the change in topology are factors that affect the choice of the protocols. There is no best protocol for all applications. There is still ongoing research on mobile ad hoc networks and the research may lead to even better protocols and will probably face new challenges.

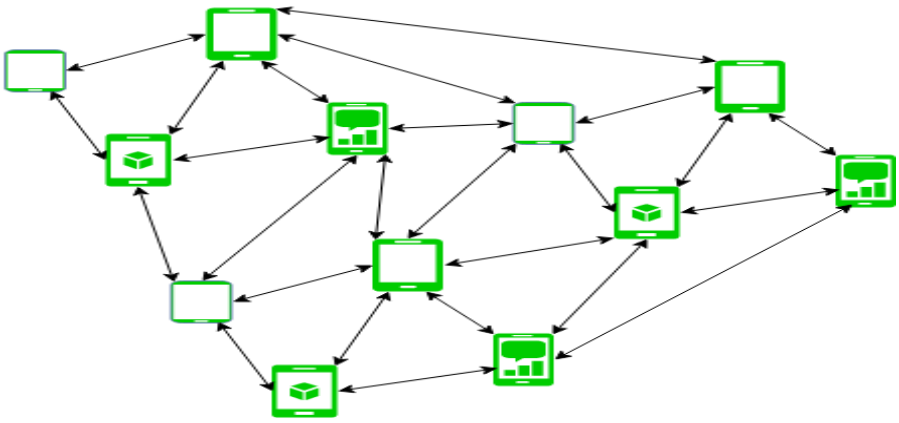


Figure - Mobile Ad Hoc Network

MANET Features (What can it do?**)**

MANET has the following features:

1. **Autonomous terminal**: In MANET, each mobile host is an autonomous node, which may function as both a host and a router. In other words, besides the basic processing ability as a host, the mobile nodes can also perform switching functions as a router. So usually endpoints and switches are indistinguishable in MANET.
2. **Distributed operation**: Since there is no background network for the central control of the network operations, the control and management of the network is distributed among the terminals. The nodes involved in a MANET should collaborate amongst themselves and each node acts as a relay as needed, to implement functions e.g. security and routing.
3. **Multi-hop routing**: Basic types of ad hoc routing algorithms can be single hop and multi-hop. Single-hop MANET is simpler than multihop in terms of structure and implementation, with the cost of lesser functionality and applicability. When delivering data packets from a source to its destination out of the direct wireless transmission range, the packets should be forwarded via one or more intermediate nodes.
4. **Dynamic network topology**: Since the nodes are mobile, the network topology may change rapidly and unpredictably and the connectivity among the terminals may vary with time. MANET should adapt to the traffic and propagation conditions as well as the mobility patterns of the mobile network nodes. The mobile nodes in the network dynamically establish routing among themselves as they move about, forming their own network on the fly. Moreover, a user in the MANET may not only operate within the ad hoc network but may require access to a public fixed network.
5. **Fluctuating link capacity**: The nature of high bit-error rates of wireless connection might be more profound in a MANET. One end-to-end path can be shared by several sessions. The channel over which the terminals communicate is subject to noise, fading, and interference, and has less bandwidth than a wired network. In some scenarios, the path between any pair of users can traverse multiple wireless links and the link themselves can be heterogeneous. One effect of the relatively low to moderate capacities is that congestion is typically the norm rather than the exception i.e. aggregate application demand will likely approach or exceed network capacity frequently.
6. **Energy-constrained operation**: Some or all of the nodes in a MANET may rely on batteries or other means for their energy. Such devices need optimized algorithms and mechanisms that implement the computing and communicating functions.

7. **Limited physical security**: MANETs are generally more prone to physical security threats than are fixed cable networks. The increased possibility of eavesdropping, spoofing and denial-of-service attacks should be carefully considered.

The following are the desirable properties of MANET routing protocol:

1. **Distributed operation**: Since there is no central point like a wired network, each and every node in the MANET performs routing.
2. **Loop-freedom:** Though, if not incorporated in the routing protocol, the Time-to-live (TTL) value could be used to prevent the packet from roaming in the network for arbitrarily long periods of time. But, still, this property is desirable for efficient use of resources and better overall performance.
3. **Demand-based operation**: Instead of assuming a uniform traffic distribution within the network and maintaining routing information between all nodes at all times, routing algorithm should adapt to the traffic pattern on a demand or need basis. It should be done in such a way so that it could utilize mobile nodes energy and network bandwidth more efficiently at the cost of increased route discovery delay.
4. **Proactive operation**: This is the opposite of demand-based operation. It certain contexts, the additional latency demand-based operation are unacceptable. If bandwidth and energy resources permit, the proactive operation is desirable in these contexts.
5. **Security**: Without some form of network-level or link layer security, a MANET routing protocol is vulnerable to many forms of attack. It may be relatively simple to snoop network traffic, manipulate packet headers and redirect routing messages, within a wireless network without appropriate security provisions.
6. **Sleep period operation**: As a result of energy conservation, or some other need to be inactive, nodes of a MANET may stop transmitting and/or receiving for arbitrary time periods. A routing protocol should be able to accommodate such sleep periods without overly adverse consequences.
7. **Unidirectional link support**: Bi-directional links are typically assumed in the design of routing algorithms, and many algorithms are incapable of functioning properly over unidirectional links. But, unidirectional links can and do occur in wireless networks.

Key Characteristics (What is it like?)

- **Infrastructure-less:** No fixed base stations or routers are required; nodes establish connections on the fly.
- **Dynamic Topology:** Nodes move freely, causing network links to break and reconnect unpredictably.
- **Multi-hop Routing:** Packets may travel through multiple intermediate nodes to reach their destination.
- **Autonomous & Self-Healing:** The network forms and repairs itself without human intervention.
- **Resource Constrained:** Typically limited by battery power,, lower bandwidth, and restricted memory.

Common Use Cases

- **Military Communications:** Battlefield communication between soldiers, vehicles, and drones.
- **Disaster Relief:** Rapidly deployed networks for rescue operations where infrastructure is damaged.
- **Vehicular Ad Hoc Networks (VANETs):** Communication between vehicles for safety and navigation.
- **Smart Home/Office:** Direct device-to-device communication without a central access point.

Advantages

- **Rapid Deployment:** Setup is fast and does not require existing infrastructure.
- **Robustness:** Self-healing, so if one node fails, others can reroute traffic.
- **Flexibility:** Enables communication in remote areas or temporary settings.

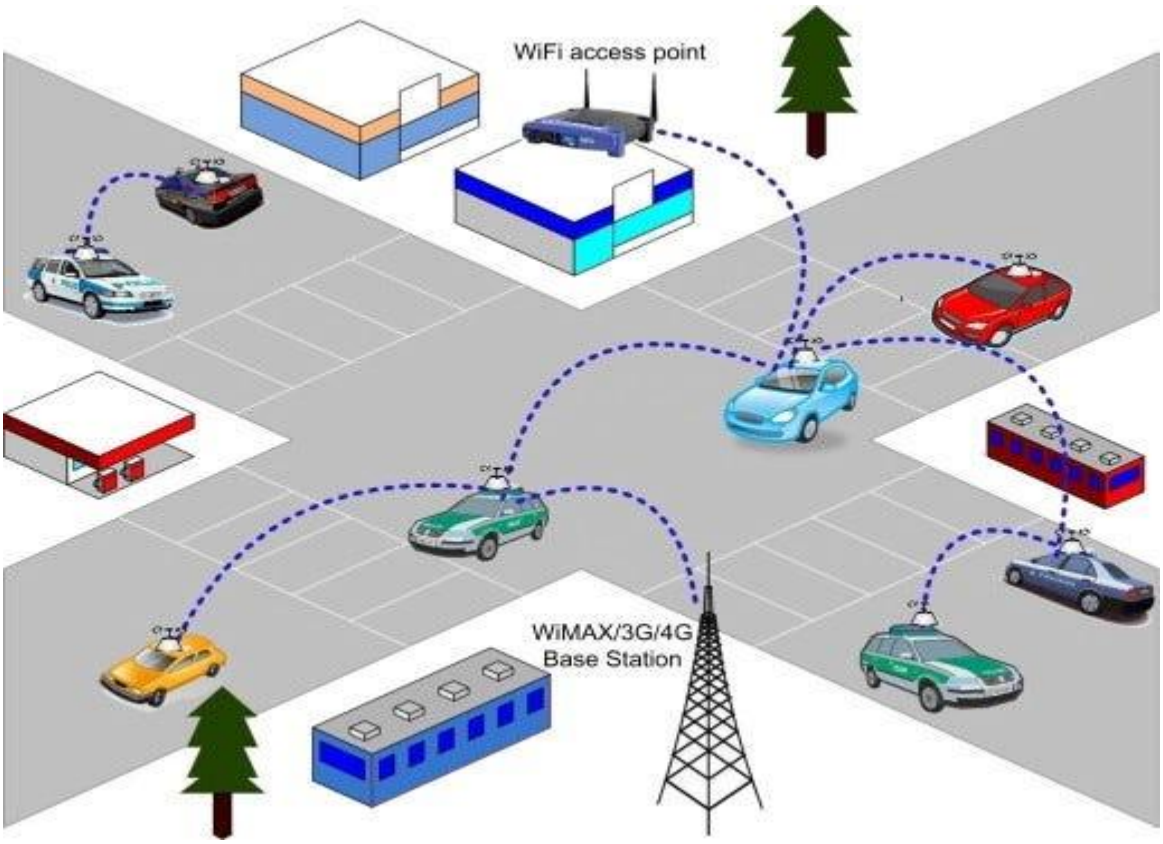
Challenges

- **Security:** High vulnerability to attacks due to open wireless medium and lack of centralized security.
- **Routing Complexity:** Maintaining efficient routing paths in a constantly changing network is difficult.
- **Energy Constraints:** Mobile nodes rely on batteries, limiting operational lifespan.
- **Limited Bandwidth:** Lower throughput compared to wired networks.

Types of MANETs

1. Vehicular Ad hoc Networks (VANETs)

Vehicular Ad hoc Networks (VANETs) are a specialized type of Mobile Ad Hoc Network (MANET) that enable wireless communication between vehicles (Vehicle-to-Vehicle, V2V) and between vehicles and roadside infrastructure (Vehicle-to-Infrastructure, V2I). Designed to improve road safety, traffic efficiency, and provide infotainment, they operate without fixed infrastructure, allowing nodes to act as both transmitters and routers.



Key Characteristics and Components

- **Components:** Vehicles are equipped with On-Board Units (OBUs), while infrastructure includes RoadSide Units (RSUs) along roads.
- **High Mobility:** Vehicles move at high speeds, leading to rapidly changing network topologies.
- **Dynamic Connectivity:** Connections are often spontaneous, short-lived, and geographically constrained.
- **Wireless Access:** Typically uses technologies like Dedicated Short-Range Communications (DSRC) or cellular networks.

Primary Applications (Intelligent Transportation Systems - ITS)

- **Safety Services:** Accident avoidance, blind-spot warnings, and emergency braking notifications.

- **Traffic Management:** Real-time traffic flow monitoring, lane guidance, and routing information.
- **Infotainment:** Internet access, location-based services, and mobile commerce on the go.

Major Challenges

- **High Mobility and Topologies:** Maintaining stable connections at high speeds is difficult.
- **Security:** Susceptible to attacks such as message tampering, unauthorized access, and privacy violations (e.g., Sybil attacks).
- **Reliability:** Ensuring consistent, low-latency communication for critical safety messages.
- **Scalability:** Managing network congestion when many vehicles are in close proximity.

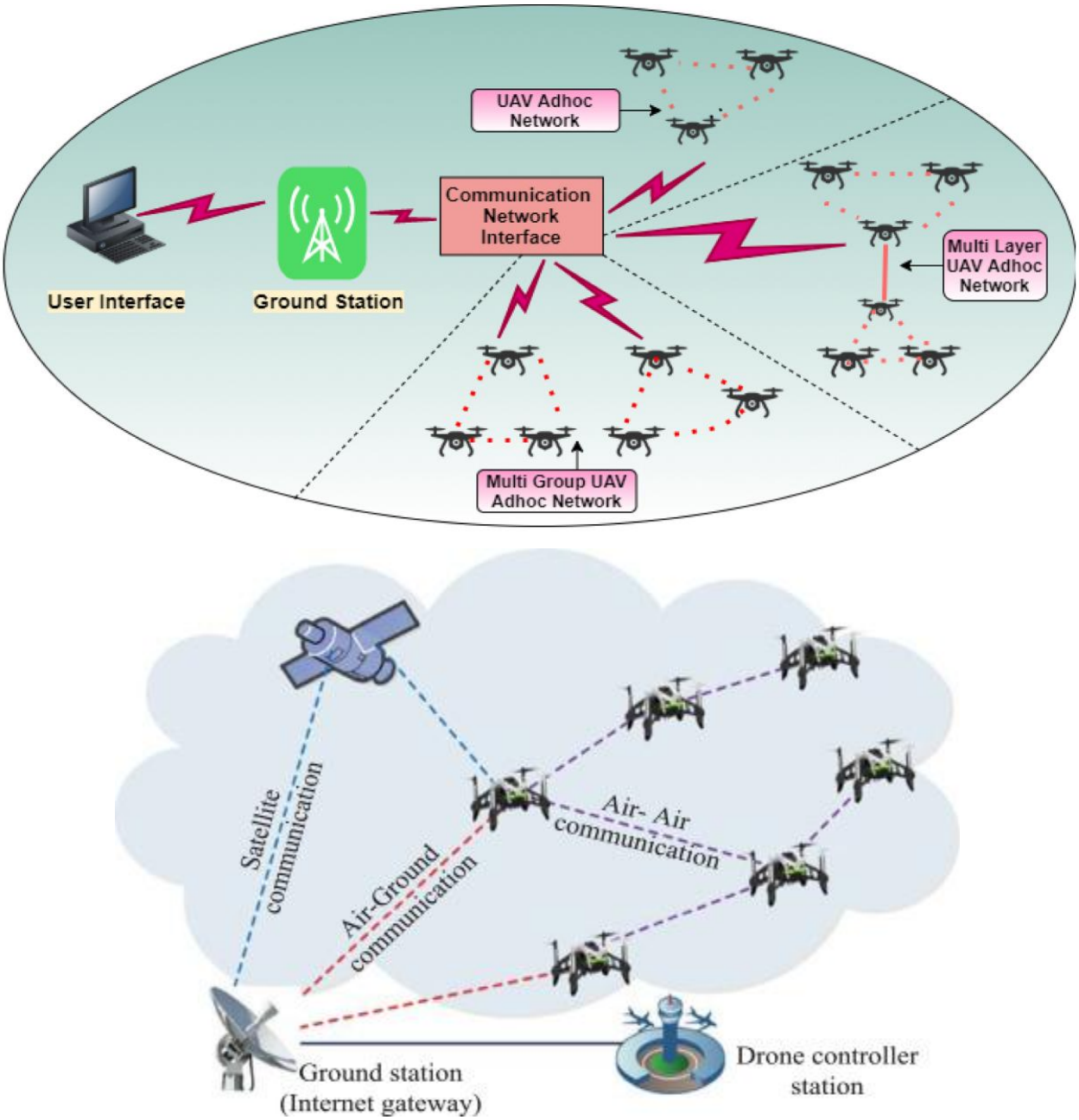
Types of Communication in VANETs

- **V2V (Vehicle-to-Vehicle):** Direct communication between cars.
- **V2I (Vehicle-to-Infrastructure):** Communication between cars and infrastructure (e.g., traffic lights, toll booths).
- **V2P (Vehicle-to-Pedestrian):** Communication between vehicles and pedestrians.

VANETs serve as a key component of Intelligent Transportation Systems (ITS) to make road transport safer and more efficient.

2. Flying Ad hoc Networks(FANETs)

Flying Ad hoc Networks (FANETs) are a specialized subclass of Mobile Ad hoc Networks (MANETs) formed by Unmanned Aerial Vehicles (UAVs or drones) that communicate and collaborate autonomously without fixed infrastructure. They enable flexible, high-speed networking for reconnaissance, monitoring, and communication, with nodes often switching roles between data collectors and communication relays.



Drone assisted Flying Ad-Hoc Networks

Key Characteristics

- **High Mobility:** Unlike VANETs or MANETs, UAVs move at high speeds, leading to very fast topology changes.
- **3D Topology:** Nodes operate in three-dimensional space, providing superior visibility and communication over wide areas.
- **Energy Constraints:** Despite better battery capacity than small ground sensors, drones still face limitations on operating time.
- **Low Node Density:** Generally fewer nodes covering a large area compared to terrestrial networks.

- **Ad Hoc Structure:** Self-organizing network setup requires no human intervention to establish, allowing for rapid deployment.

Key Applications

- **Search and Rescue (SAR):** Rapid mapping of disaster areas and detection of survivors.
- **Military Surveillance:** UAV swarms collaborating for battlefield monitoring.
- **Precision Agriculture:** Automated monitoring of crops and environmental analysis.
- **Infrastructure Inspection:** Monitoring pipelines, power lines, and construction sites.

Challenges

- **Routing Complexity:** Frequently changing links make it hard to maintain consistent routes, requiring robust routing protocols.
- **Limited Power:** High-performance drone operations are constrained by battery life.
- **Security Vulnerabilities:** Open wireless communications make them susceptible to attacks, including signal jamming and spoofing.
- **Localization:** Precise, fast positioning of nodes is essential for network management.

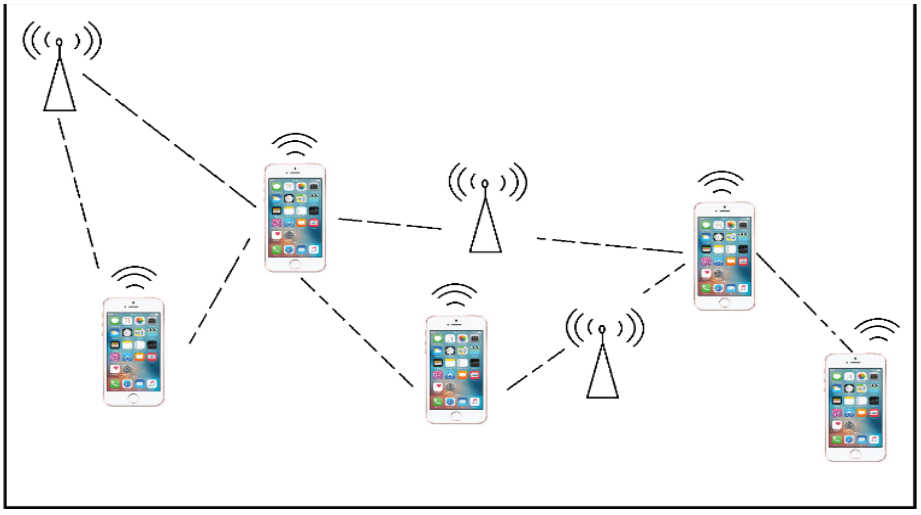
Routing Protocols in FANETs

- **Topology-based:** Uses network link information to route packets (e.g., proactive or reactive).
- **Position-based:** Uses GPS information to direct packets to a physical location.
- **Cluster-based:** Groups UAVs to improve scalability and resource management.

FANETs represent a high-growth area in networking, combining mobile communication with robotics, often considered for integration with 5G and 6G systems for better reliability.

3. Smartphone Ad hoc Networks(SPANs)

Smartphone Ad hoc Networks (SPANs) are decentralized, self-configuring peer-to-peer networks created by mobile devices—primarily smartphones—using Bluetooth or Wi-Fi without relying on cellular carriers or central access points. SPANs allow devices in close proximity to share data, enabling free communication and file sharing when network infrastructure is overloaded, unavailable, or nonexistent.



- Each smartphone connects directly to nearby phones.
- No Wi-Fi router or mobile tower is needed.
- Messages travel from one phone to another (multi-hop).

Key Characteristics & Capabilities

- **Peer-to-Peer Networking:** Direct device-to-device connection (e.g., Bluetooth/Wi-Fi direct).
- **Multi-hop Routing:** Data can travel through intermediate phones to reach a destination outside direct transmission range, unlike simple hub-and-spoke networks.
- **Infrastructure-less:** Operates independently of base stations or internet access.
- **Self-Organizing:** Nodes (phones) can join or leave the network voluntarily without breaking functionality.

Key Advantages

- **Cost-Free Data Exchange:** Enables direct messaging and file sharing without using data plans, popular for local communications.
- **Disaster Recovery:** Critical in emergencies where cellular towers are destroyed or overloaded.
- **High Mobility:** Suitable for crowd scenarios like music festivals, conferences, or large public gatherings.

Common Use Cases

- **Localized Communication:** Sharing media, files, or messages at events without cellular network access.
- **Community Networking:** Providing connectivity in rural or underdeveloped areas.
- **Crowd-sourced Sensing:** Utilizing smartphone sensors to gather data within a local area.

Differences from Other Technologies

- Unlike **Wi-Fi Direct**, SPANs often support more complex multi-hop networking rather than just 1-to-1 or simple 1-to-many connections.
- Unlike **MANETs (general)**, which may use specialized radio equipment, SPANs are specifically focused on leveraging the Bluetooth and Wi-Fi hardware inside standard smartphones.

Challenges

- **Security:** Susceptible to attacks due to the open nature of wireless communication.
- **Energy Consumption:** Intensive use of wireless radios can drain smartphone batteries rapidly.
- **Routing Complexity:** Managing constantly shifting connections and routes in a dynamic group.

Routing

Routing is the process of finding the best path for traffic in a network, or across multiple networks. The role of routing is similar to the road map for a hotel. In both cases, we need to deliver messages at proper location and in an appropriate way.

Routing in a mobile ad-hoc network depends on many factors such as:

- Modeling of the topology,
- Selection of routers,
- Initiation of a route request,
- And specific underlying characteristics that could serve as heuristics in finding the path effectively.

Need for Routing

- Since centralized routing in a dynamic and even for small networks is impossible therefore routing computation must be distributed.
- Route computation should not add many more nodes.
- If any host demands for the route, they must have quick access.
- Maintenance of a global state should not involve in the route computation.
- Each node should care about their destination node to its route and should not be involved in frequent topology updates for those portions of the network that have no traffic.

- Since broadcast can be time consuming for MANETs, it must be avoided as much as possible.
- In routing there must have a backup route when the primary route has become stale.

Routing Classification

Routing protocol can be classified as:

1. Proactive Protocol
2. Reactive Protocol
3. Hybrid Protocol
4. Hierarchical Protocol
5. Geographic Protocol

1. Proactive Protocol

Proactive protocols attempt to evaluate continuously the routes within the network. It means proactive protocol continuously maintain the routing information, so that when a packet needs to be forwarded, the path is known already and can be immediately used.

The advantage of the proactive schemes is that whenever a route is needed, there is negligible delay in determining the route.

Unfortunately, it is a big overhead to maintain routing tables in the MANET environment. Therefore, this type of protocol has following common disadvantages:

- Requires more amounts of data for maintaining routing information.
- Low reaction on re-structuring network and failures of individual nodes.

2. Reactive Protocols

Reactive protocols do not maintain routes but invoke a route determination procedure only on demand or we can say reactive protocols build the routes only on demand. Thus, when a route is required, some sort of global search procedure is initiated.

These protocols have the following advantages:

- No large overhead for global routing table maintenance as in proactive protocols.
- Reaction is quick for network restructure and node failure.

  Even though reactive protocols have become the main stream for MANET routing, they still have the following disadvantages:
- Latency time is high in route finding
- Excessive flooding can lead to network clogging.

3. Hybrid Protocols

Hybrid protocols attempt to take advantage of best of reactive and proactive schemes. The basic idea behind such protocols is to initiate route discovery on demand but at a limited search cost. One of the popular hybrid protocols is zone routing protocol (ZRP).

4.Hierarchical Routing Protocols

With this type of protocol, the choice of proactive and of reactive routing depends on the hierarchical level in which a node resides. The routing is initially established with some proactively prospected routes and then serves the demand from additionally activated nodes through reactive flooding on the lower levels. The choice for one or the other method requires proper attribution for respective levels. The main disadvantages of such algorithms are:

1. Advantage depends on depth of nesting and addressing scheme.
2. Reaction to traffic demand depends on meshing parameters.

Routing protocols may also be categorized as follows:

1. Table-driven protocols
2. Source initiated on -demand protocols

1. Table-driven routing protocol

- These protocols are called table-driven because each node is required to maintain one or more tables containing routing information on every other node in the network.
- They are **proactive** in nature so that the routing information is always consistent and up to date.
- The protocols respond to changes in network topology by propagating the updates throughput the network so that every node has a consistent view of the network.

The table-driven routing protocols are categorized as: Destination - sequenced distance vector routing, Cluster Head gateway switch Routing, Wireless routing protocol (WRP)

2. Source initiated on -demand protocols

- Source - initiated on demand routing is **reactive** in nature, unlike table driven routing. This type of protocols generates routes only when a source demands it.

- In other words, when a source node requires a route to a destination, the source initiates a route discovery process in the network. This process finishes when a route to the destination has been discovered or all possible routes have been examined without any success.
- The discovered route is maintained by a route maintenance procedure, until it is no longer desired or the destination becomes inaccessible.

The source initiated on demand routing is categorized as follows: Ad hoc on demand distance vector routing (AODV), Dynamic Source Routing (DSR)

Assignment

- **Explain the concept of "Ad-hoc Networks" in mobile computing.**
- **Distinguish between infrastructured and infrastructureless wireless networks**